TEACHER ANSWER KEY

FTC StarterBot TeleOp Answer Key

Part 8 Teacher Reference

Use with the Part 8 student workbook and corrected StarterBot program

This key supplies expected calculations and reasoning. Accept equivalent student language when the explanation preserves the code behavior and the safety requirement.

Before the video

- Drive the chassis
- Run the intake roller
- Move both corner feeders

Vocabulary

TERM	EXPECTED MEANING
iterative OpMode	An OpMode whose lifecycle methods are called by the FTC runtime; loop repeats after START.
hardwareMap	The FTC lookup that connects a quoted configuration name to a Java hardware object.
split arcade	A control style where left-stick Y drives forward and backward while right-stick X rotates.
CRServo	A continuous-rotation servo controlled by signed power for speed and direction.
telemetry	Data sent to the Driver Station for display and diagnosis.

Lifecycle order

- 1. The runtime calls init.
- 2. hardwareMap connects the devices.
- 3. The team presses START.
- 4. loop repeats and reads the controls.
- 5. The team presses STOP.
- 6. The runtime calls stop.

Direction BRAKE and safety

The right motor is reversed because mirrored axles point in opposite physical directions; opposite software directions can make both sides agree on forward.

BRAKE reduces coasting when zero power is commanded. It does not mechanically lock the robot.

Stopped wheel motion is not enough because the OpMode may still be active and a new command can arrive. Press STOP and confirm DISABLED before approach.

Split arcade drive answers

CASE	FORWARD	ROTATE	LEFT	RIGHT	EXPECTED MOTION
Stopped	0.00	0.00	0.00	0.00	No drivetrain motion
Forward	+0.60	0.00	+0.60	+0.60	Straight forward
Reverse	-0.50	0.00	-0.50	-0.50	Straight reverse
Turn right	0.00	+0.40	+0.40	-0.40	Rotate right
Forward right	+0.60	+0.20	+0.80	+0.40	Curve right
Full mix	+1.00	+1.00	+2.00	0.00	Normalize before sending

Normalization challenge

For forward = +1.00 and rotate = +1.00: raw left = +2.00 and raw right = 0.00. The maximum magnitude is 2.00. Normalized left = +1.00 and normalized right = 0.00.

Both sides are divided by the same number so the left-to-right power ratio and intended steering relationship are preserved.

Explain the curve

For forward = +0.60 and rotate = +0.20, the left side receives +0.80 and the right side receives +0.40. Both sides move forward, but the left side is faster, so the chassis follows a curve rather than rotating in place.

Intake trigger answers

CASE	RIGHT	LEFT	INTAKE POWER	EXPECTED RESULT
Released	0.00	0.00	0.00	Stopped

CASE	RIGHT	LEFT	INTAKE POWER	EXPECTED RESULT
Intake in	1.00	0.00	+1.00	Roller and feeders inward
Intake out	0.00	1.00	-1.00	Roller and feeders reverse
Half speed in	0.50	0.00	+0.50	Half-speed inward
Equal triggers	0.75	0.75	0.00	Stopped

The three devices are the intake roller motor, left intake CRServo, and right intake CRServo. The right servo is reversed so mirrored feeders rotate in opposite numeric directions but pull material inward together. CRServo power controls speed and direction, not an exact position.

Hardware map answers

JAVA VARIABLE	DEVICE TYPE	CONFIGURED NAME	PHYSICAL JOB
leftDrive	DcMotor	left_drive	Left drivetrain motor
rightDrive	DcMotor	right_drive	Right drivetrain motor
intake	DcMotorEx	intake	Intake roller motor
leftIntakeServo	CRServo	left_intake_servo	Left corner feeder
rightIntakeServo	CRServo	right_intake_servo	Right corner feeder

Wheels up expected observations

TEST	EXPECTED EVIDENCE
Left drive	Only the intended left side moves; direction matches the command.
Right drive	Only the intended right side moves; direction matches the command.
Straight forward	Both sides agree on physical forward.
Rotate	The two sides oppose each other in the intended direction.
Intake inward	Roller and both feeder paddles move material inward.
Intake outward	All three intake devices reverse together.

TEST	EXPECTED EVIDENCE
Equal triggers	Calculated intakePower is zero and intake devices stop.
STOP and DISABLED	Outputs stop and the Driver Station visibly confirms DISABLED.

Troubleshooting evidence order

- Read the calculation or setPower command in the code.
- Confirm the quoted name and configured physical device.
- Compare the reported input and power values in telemetry.
- Observe the mechanism from outside the boundary.
- Choose one hypothesis that fits all evidence.
- Press STOP, confirm DISABLED, make one change, and repeat the controlled test.

Exit response criteria

A complete response identifies the drive sticks and equations, the trigger-subtraction intake command, all three intake devices, and at least one valid safety or diagnostic check. Strong responses explain that telemetry reports data but does not prove physical motion.